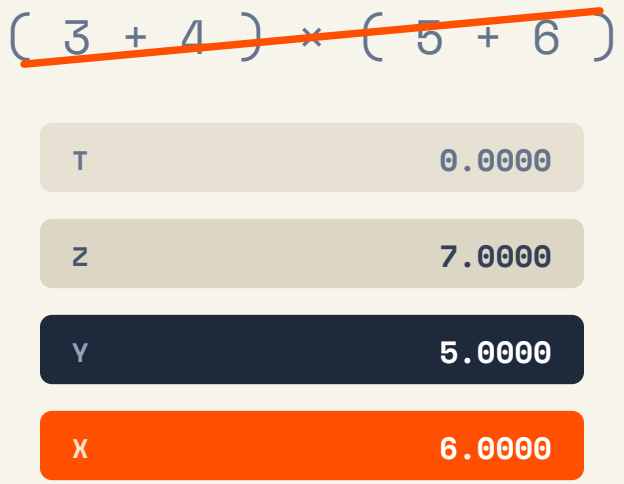
STUDENT FLUENCY & COMPETITION WORKBOOK • TRIALS 1–12 & C1–C3

STACK SPEED & FLUENCY

15 Timed Number Sense & RPN Proficiency Sprints



StackCalc

COMPETITION RULES & COMPETITOR LOG

Stack Speed & Fluency Trials

UIL-style timed RPN speed and accuracy sprints across Stack Depth Levels 1, 2, and 4.

1. INVITATIONAL TIME TRIAL & SPEED SCORING RULES

Modeled after University Interscholastic League (UIL) Calculator Applications competitions, these 15 timed sprints build rapid, inside-out fluency with Reverse Polish Notation (RPN) on the SC-6 Elementary (Trials 1–7, C1) and SC-32 Scientific (Trials 8–12, C2–C3) calculators.

CORRECT ANSWER	INCORRECT / ERROR	UNATTEMPTED / SKIP	SPEED BONUS
+5 Points	-2 Points	0 Points	+1 pt / rem. min

2. MAXIMUM STACK LEVEL DISCIPLINE

Competitors must respect the Max Stack Depth badge on each trial: Level 1 uses only Register X; Level 2 uses Registers X and Y; Level 4 uses the full 4-register stack (T, Z, Y, X). Pressing unnecessary [ENTER] keys that overflow the allowed depth is penalized.

3. OFFICIAL COMPETITOR TIME TRIAL LOG SHEET

Trial	Mathematical Topic	Max Stack	Time	Attempt	Correct	Errors	
T-1	Unit 1 Time Trial: Step Counter & Accu	Level 1	5 min	___/10	___(+5)	___(-2)	_____
T-2	Unit 2 Time Trial: Part-Part-Whole & S	Level 2	5 min	___/10	___(+5)	___(-2)	_____
T-3	Unit 3 Time Trial: Ten-Frames & Comple	Level 2	5 min	___/10	___(+5)	___(-2)	_____
T-4	Unit 4 Time Trial: Number Racks, Doubl	Level 2	5 min	___/10	___(+5)	___(-2)	_____
T-CUM_1	Cumulative Trial C1: Primary Stack Fun	Level 2	10 min	___/10	___(+5)	___(-2)	_____
T-5	Unit 5 Time Trial: 4-Register Stack &	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-6	Unit 6 Time Trial: Fraction Studio & M	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-7	Unit 7 Time Trial: Postfix Evaluation	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-8	Unit 8 Time Trial: Tactile Algebra & Q	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-CUM_2	Cumulative Trial C2: Intermediate Stac	Level 4	15 min	___/10	___(+5)	___(-2)	_____
T-9	Unit 9 Time Trial: Trigonometry, Vecto	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-10	Unit 10 Time Trial: Exponential Growth	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-11	Unit 11 Time Trial: Combinatorics, Sta	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-12	Unit 12 Time Trial: Digital Logic, Bas	Level 4	10 min	___/10	___(+5)	___(-2)	_____
T-CUM_3	Cumulative Trial C3: Grand RPN Invitat	Level 4	20 min	___/10	___(+5)	___(-2)	_____

How to Run Weekly 5-Minute Friday Sprints

Start the timer when students turn the page to the week's Trial. Students solve each problem on their SC-6 or SC-32 calculator, record their keystroke plan and final answer, and check their score against the Self-Check Key (pp. 17–18).

Unit 1 Time Trial: Step Counter & Accumulator Sprint

Foundations of Counting, Cardinality & The One-More Step Counter • Focus: Ten-Frame Tray & Counters

MAX STACK DEPTH: 1 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**1 + 1 + 1 (Count 3 steps)** Stack Hint: Single register accumulator	Keystroke Plan:	**Final Answer:**
2	**5 + 1 + 1 (Start at 5, step 2)** Stack Hint: Accumulate in X	Keystroke Plan:	**Final Answer:**
3	**Step count 4 items: start 0, step +1** Stack Hint: Repeat + key	Keystroke Plan:	**Final Answer:**
4	**Count 8 items with auto-lift** Stack Hint: Continuous + entry	Keystroke Plan:	**Final Answer:**
5	**Count down: start 6, step -1, step -1** Stack Hint: Subtract step	Keystroke Plan:	**Final Answer:**
6	**Clear and count: C then 1+1+1+1+1** Stack Hint: Clear before count	Keystroke Plan:	**Final Answer:**
7	**Count by 2s: 2 + 2 + 2** Stack Hint: Skip count by 2	Keystroke Plan:	**Final Answer:**
8	**Step count to ten: 7 + 1 + 1 + 1** Stack Hint: Reach upper frame	Keystroke Plan:	**Final Answer:**
9	**Zero set: 0 + 9** Stack Hint: Identity of zero	Keystroke Plan:	**Final Answer:**
10	**Count back from 10: 10 - 1 - 1 - 1** Stack Hint: Decrement from 10	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 1 on this sprint. Round non-exact decimals to 4 significant digits.

Unit 2 Time Trial: Part-Part-Whole & Stack Addition

Part-Part-Whole Domino Decompositions via Stack Registers Y and X • Focus: Math Domino Studio (12-Frame Tray)

MAX STACK DEPTH: 2 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Part 4, Part 3 -> Whole?** Stack Hint: Y=4, X=3, [+] combines	Keystroke Plan:	**Final Answer:**
2	**Part 5, Part 5 -> Whole?** Stack Hint: Ten bond	Keystroke Plan:	**Final Answer:**
3	**Whole 9, Part 6 -> Missing Part?** Stack Hint: Decompose whole 9	Keystroke Plan:	**Final Answer:**
4	**Whole 10, Part 4 -> Missing Part?** Stack Hint: Decompose whole 10	Keystroke Plan:	**Final Answer:**
5	**Part 8, Part 7 -> Whole?** Stack Hint: Sum past ten	Keystroke Plan:	**Final Answer:**
6	**Whole 14, Part 8 -> Missing Part?** Stack Hint: Decompose 14	Keystroke Plan:	**Final Answer:**
7	**Part 6, Part 7 -> Whole?** Stack Hint: Part combination	Keystroke Plan:	**Final Answer:**
8	**Whole 18, Part 9 -> Missing Part?** Stack Hint: Double decomposition	Keystroke Plan:	**Final Answer:**
9	**Part 12, Part 8 -> Whole?** Stack Hint: Full twenty bond	Keystroke Plan:	**Final Answer:**
10	**Whole 25, Part 11 -> Missing Part?** Stack Hint: Two-digit decompose	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this sprint. Round non-exact decimals to 4 significant digits.

Unit 3 Time Trial: Ten-Frames & Complements to 10

5-Structure, Complements & Bridging Through 10 • Focus: Ten-Frame Tray & Counters

MAX STACK DEPTH: 2 | TIME LIMIT: 5 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ____:____ Score: ____ / 50

1	**Complement of 7 to 10** Stack Hint: 10 in Y, 7 in X	Keystroke Plan:	Final Answer:
2	**Complement of 4 to 10** Stack Hint: 10 in Y, 4 in X	Keystroke Plan:	Final Answer:
3	**Bridge: 8 + 5 = 8 + 2 + 3** Stack Hint: Bridge through 10	Keystroke Plan:	Final Answer:
4	**Bridge: 9 + 6 = 9 + 1 + 5** Stack Hint: Bridge through 10	Keystroke Plan:	Final Answer:
5	**Bridge: 7 + 6 = 7 + 3 + 3** Stack Hint: Bridge through 10	Keystroke Plan:	Final Answer:
6	**Complement of 1 to 10** Stack Hint: 10 in Y, 1 in X	Keystroke Plan:	Final Answer:
7	**Double frame: 20 - 7** Stack Hint: Subtract across 2 frames	Keystroke Plan:	Final Answer:
8	**Bridge: 8 + 7 = 8 + 2 + 5** Stack Hint: Bridge through 10	Keystroke Plan:	Final Answer:
9	**Place value: 10 + 6** Stack Hint: Teen number formation	Keystroke Plan:	Final Answer:
10	**Frame difference: 10 - 2** Stack Hint: Ten complement	Keystroke Plan:	Final Answer:

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this sprint. Round non-exact decimals to 4 significant digits.

TIMED SPEED & FLUENCY SPRINT • SC-6 ELEMENTARY

Unit 4 Time Trial: Number Racks, Doubles & Groups

5-Structure, Automatic Lift Duplication & Early Multiplication • Focus: Two-Row Number Rack (Rekenrek)

MAX STACK DEPTH: 2 | TIME LIMIT: 5 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Double 6: 6 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Plan:	**Final Answer:**
2	**Double 8: 8 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Plan:	**Final Answer:**
3	**Near-Double: 7 + 8** Stack Hint: Double 7 plus 1	Keystroke Plan:	**Final Answer:**
4	**Near-Double: 6 + 7** Stack Hint: Double 6 plus 1	Keystroke Plan:	**Final Answer:**
5	**Double 9: 9 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Plan:	**Final Answer:**
6	**Near-Double: 8 + 9** Stack Hint: Double 8 plus 1	Keystroke Plan:	**Final Answer:**
7	**Equal groups: 3 groups of 4** Stack Hint: Multiply group size	Keystroke Plan:	**Final Answer:**
8	**Equal groups: 4 groups of 5** Stack Hint: Multiply group size	Keystroke Plan:	**Final Answer:**
9	**Double 12: 12 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Plan:	**Final Answer:**
10	**Near-Double: 9 + 10** Stack Hint: Double 9 plus 1	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this sprint. Round non-exact decimals to 4 significant digits.

Cumulative Trial C1: Primary Stack Fundamentals Championship

Comprehensive Time Trial across Units 1–4 (Max Stack Depth 2) • Focus: Primary Math Manipulative Lab

MAX STACK DEPTH: 2 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Step count 7 items from 1** Stack Hint: Accumulate in X	Keystroke Plan:	**Final Answer:**
2	**Number bond: 9 and 7 -> Whole?** Stack Hint: Y=9, X=7	Keystroke Plan:	**Final Answer:**
3	**Decompose: Whole 17, Part 8** Stack Hint: 17 in Y, 8 in X	Keystroke Plan:	**Final Answer:**
4	**Complement of 3 to 10** Stack Hint: Ten complement	Keystroke Plan:	**Final Answer:**
5	**Double 14: 14 ENTER +** Stack Hint: Duplicate to Y	Keystroke Plan:	**Final Answer:**
6	**Near-double: 11 + 12** Stack Hint: Double 11 plus 1	Keystroke Plan:	**Final Answer:**
7	**Bridge through 10: 9 + 8** Stack Hint: Bridge with frame	Keystroke Plan:	**Final Answer:**
8	**Equal groups: 6 groups of 5** Stack Hint: 5 in Y, 6 in X	Keystroke Plan:	**Final Answer:**
9	**Decompose: Whole 30, Part 12** Stack Hint: Missing part	Keystroke Plan:	**Final Answer:**
10	**Combined: Double 8 then add 5** Stack Hint: Chained operations	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this sprint. Round non-exact decimals to 4 significant digits.

Unit 5 Time Trial: 4-Register Stack & Four Operations

T-Z-Y-X Flow, Automatic Lift & Parenthesis-Free Arithmetic • Focus: SC-32 Hardware LCD Stack & Place Value Studio

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**(18 - 6) × 3** Stack Hint: Auto-lift pushes 12 to Y	Keystroke Plan:	**Final Answer:**
2	**(24 + 16) ÷ 8** Stack Hint: Sum in Y, divide by 8	Keystroke Plan:	**Final Answer:**
3	**45 ÷ (12 - 3)** Stack Hint: Push 45 to Y, sub in X	Keystroke Plan:	**Final Answer:**
4	**(15 × 4) + (18 ÷ 2)** Stack Hint: Two branch accumulation	Keystroke Plan:	**Final Answer:**
5	**(50 - 14) ÷ (3 + 6)** Stack Hint: Stack holds numerator in Y	Keystroke Plan:	**Final Answer:**
6	**12 × 5 - 8 × 4** Stack Hint: Two branch subtraction	Keystroke Plan:	**Final Answer:**
7	**((20 - 8) × 4) ÷ 6** Stack Hint: Chained pipeline	Keystroke Plan:	**Final Answer:**
8	**72 ÷ (4 + 5) × 3** Stack Hint: Parenthesis-free stack	Keystroke Plan:	**Final Answer:**
9	**100 - (16 × 5 - 12)** Stack Hint: Deep stack evaluate	Keystroke Plan:	**Final Answer:**
10	**(8 + 4) × (9 - 3) ÷ 12** Stack Hint: Multi-step evaluate	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

TIMED SPEED & FLUENCY SPRINT • SC-6 ELEMENTARY

Unit 6 Time Trial: Fraction Studio & Mixed Numbers

Two-Dot Notation, Reciprocals & Height Equivalence • Focus: 84 mm Fraction Towers & Comparator Well

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**1/2 + 1/4 (Two-dot entry)** Stack Hint: Decimal equivalent 0.75	Keystroke Plan:	**Final Answer:**
2	**3/4 - 1/2** Stack Hint: Decimal equivalent 0.25	Keystroke Plan:	**Final Answer:**
3	**Mixed: 2 1/2 + 1 1/4** Stack Hint: Mixed number addition	Keystroke Plan:	**Final Answer:**
4	**Reciprocal of 4/5: .4.5 [1/x]** Stack Hint: Reciprocal key	Keystroke Plan:	**Final Answer:**
5	**3/8 + 1/4** Stack Hint: Three eighths plus fourth	Keystroke Plan:	**Final Answer:**
6	**2/3 × 3/4** Stack Hint: Fraction multiplication	Keystroke Plan:	**Final Answer:**
7	**3/4 ÷ 1/2** Stack Hint: Fraction division	Keystroke Plan:	**Final Answer:**
8	**1 1/2 × 4** Stack Hint: Mixed scaling	Keystroke Plan:	**Final Answer:**
9	**5 - 2 3/4** Stack Hint: Subtract mixed number	Keystroke Plan:	**Final Answer:**
10	**(1/3 + 1/6) × 12** Stack Hint: Combined fraction pipeline	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

Unit 7 Time Trial: Postfix Evaluation & Chained Operations

Postfix Evaluation, Stack Registers & Parenthesis Elimination • Focus: Counting & Array Tiles + 4-Register Stack

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	(8 + 2) × (9 - 4) Stack Hint: Left-Right-Multiply	Keystroke Plan:	Final Answer:
2	((14 - 6) × 3) + (24 ÷ 4) Stack Hint: Two subtree combine	Keystroke Plan:	Final Answer:
3	(30 - (4 × 5)) × (12 ÷ (2 + 4)) Stack Hint: Deep nested subtrees	Keystroke Plan:	Final Answer:
4	(5 + 3) × (8 - 2) ÷ (6 - 2) Stack Hint: Three branch traversal	Keystroke Plan:	Final Answer:
5	((18 ÷ 3) + 4) × ((15 - 7) ÷ 2) Stack Hint: Postorder multiplication	Keystroke Plan:	Final Answer:
6	(10 + 5 × 2) ÷ (12 - 2 × 4) Stack Hint: Eliminating PEMDAS ambiguity	Keystroke Plan:	Final Answer:
7	(25 - 5 × 3) × (4 + 6 ÷ 2) Stack Hint: Nested tree traversal	Keystroke Plan:	Final Answer:
8	((7 + 3) × 4 - 8) ÷ 4 Stack Hint: Chained operations	Keystroke Plan:	Final Answer:
9	(48 ÷ (2 + 6)) × (15 - 3 × 3) Stack Hint: Tree sub-expression	Keystroke Plan:	Final Answer:
10	(16 - (4 + 2)) × (18 ÷ (3 × 2)) Stack Hint: Stack tree depth 4	Keystroke Plan:	Final Answer:

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

Unit 8 Time Trial: Tactile Algebra & Quadratic Factoring

Area Modeling, Zero-Pair Trinomials & [POLY] Solver • Focus: Algebra Area Tiles & Factoring Frame

MAX STACK DEPTH: 4 | TIME LIMIT: 10 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Factor $x^2 + 5x + 6$ via [POLY]** Stack Hint: Roots -2, -3; factors (x+2)(x+3)	Keystroke Plan:	**Final Answer:**
2	**Factor $x^2 + 7x + 12$ via [POLY]** Stack Hint: Roots -3, -4; factors (x+3)(x+4)	Keystroke Plan:	**Final Answer:**
3	**Factor $x^2 + 8x + 15$ via [POLY]** Stack Hint: Roots -3, -5; factors (x+3)(x+5)	Keystroke Plan:	**Final Answer:**
4	**Factor $x^2 - x - 6$ via [POLY]** Stack Hint: Roots 3, -2; factors (x-3)(x+2)	Keystroke Plan:	**Final Answer:**
5	**Factor $x^2 - 5x + 6$ via [POLY]** Stack Hint: Roots 2, 3; factors (x-2)(x-3)	Keystroke Plan:	**Final Answer:**
6	**Difference of squares: $x^2 - 9$** Stack Hint: Roots 3, -3; factors (x-3)(x+3)	Keystroke Plan:	**Final Answer:**
7	**Difference of squares: $x^2 - 16$** Stack Hint: Roots 4, -4; factors (x-4)(x+4)	Keystroke Plan:	**Final Answer:**
8	**Polynomial at x=3: $2x^2 + 4x - 5$** Stack Hint: Quadratic evaluation	Keystroke Plan:	**Final Answer:**
9	**Factor $2x^2 + 5x + 2$ via [POLY]** Stack Hint: Roots -0.5, -2; (2x+1)(x+2)	Keystroke Plan:	**Final Answer:**
10	**Double root: $x^2 + 10x + 25$** Stack Hint: Double root -5; $(x+5)^2$	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

Cumulative Trial C2: Intermediate Stack Mastery Championship

Comprehensive Time Trial across Units 1–8 (Max Stack Depth 4) • Focus: SC-32 Hardware LCD Stack & Manipulative Lab

MAX STACK DEPTH: 4 | **TIME LIMIT: 15 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**(32 - 14) × 5 ÷ 6** Stack Hint: Auto-lift chaining	Keystroke Plan:	**Final Answer:**
2	**3/4 + 1/2 × 1/4 (order of ops)** Stack Hint: Fractions on stack	Keystroke Plan:	**Final Answer:**
3	**Distance: P1(3, 4) to P2(9, 12)** Stack Hint: Pythagorean distance	Keystroke Plan:	**Final Answer:**
4	**Mixed: 3 1/2 - 1 3/4** Stack Hint: Two-dot fraction subtraction	Keystroke Plan:	**Final Answer:**
5	**Postfix evaluation: ((15+5)×(8-3))÷10** Stack Hint: Postfix evaluation	Keystroke Plan:	**Final Answer:**
6	**Double 18 then sub 9** Stack Hint: Duplicate to Y	Keystroke Plan:	**Final Answer:**
7	**Slope: P1(-1, -2) to P2(3, 10)** Stack Hint: Negative coordinates	Keystroke Plan:	**Final Answer:**
8	**Four operations: 80 - 15 × 4 + 12** Stack Hint: RPN stack evaluation	Keystroke Plan:	**Final Answer:**
9	**Reciprocal sum: 1/(1/4 + 1/6)** Stack Hint: Reciprocal pipeline	Keystroke Plan:	**Final Answer:**
10	**Distance from origin to (12, 16)** Stack Hint: Radial distance	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

Unit 9 Time Trial: Trigonometry, Vectors & Polar Pipeline

Right Triangles, Polar Conversions & Vector Decomposition • Focus: Unit Circle Trigonometry Volvelle

MAX STACK DEPTH: 4 | TIME LIMIT: 10 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ____:____ Score: ____ / 50

		Keystroke Plan:	Final Answer:
1	**sin(30°)** Stack Hint: Basic sine ratio		
2	**cos(60°)** Stack Hint: Basic cosine ratio		
3	**tan(45°)** Stack Hint: Tangent ratio		
4	**Hypotenuse 10, angle 30° -> Opposite (y)** Stack Hint: Opposite = h × sin(theta)		
5	**Adjacent (x): Hypotenuse 12, angle 60°** Stack Hint: Adjacent = h × cos(theta)		
6	**Polar to Rect: r=10, theta=0° -> x** Stack Hint: Polar conversion key		
7	**Rect to Polar: x=3, y=4 -> magnitude r** Stack Hint: Vector magnitude		
8	**Rect to Polar: x=5, y=12 -> magnitude r** Stack Hint: Vector magnitude		
9	**Vector magnitude: components 8 and 6** Stack Hint: Polar magnitude		
10	**asin(0.5) in degrees** Stack Hint: Inverse sine		

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

TIMED SPEED & FLUENCY SPRINT • SC-32 SCIENTIFIC

Unit 10 Time Trial: Exponential Growth & Logarithms

Natural Base e, Continuous Compounding & Doubling Time • Focus: Fraction Studio & SC-32 Hardware LCD Stack

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**2^5 (Power key)** Stack Hint: y to power x	Keystroke Plan:	**Final Answer:**
2	**3^4 (Power key)** Stack Hint: y to power x	Keystroke Plan:	**Final Answer:**
3	**ln(e)** Stack Hint: Natural log inverse	Keystroke Plan:	**Final Answer:**
4	**e^0** Stack Hint: Base e exponent	Keystroke Plan:	**Final Answer:**
5	**log10(1000)** Stack Hint: Common log	Keystroke Plan:	**Final Answer:**
6	**Compound: 1000 × (1.05)^2** Stack Hint: Annual compounding	Keystroke Plan:	**Final Answer:**
7	**Doubling time: ln(2) ÷ 0.07** Stack Hint: Continuous doubling rule	Keystroke Plan:	**Final Answer:**
8	**Half-life: 100 × (0.5)^3** Stack Hint: Decay model	Keystroke Plan:	**Final Answer:**
9	**Continuous: 500 × e^(0.04 × 5)** Stack Hint: P e^(rt) formula	Keystroke Plan:	**Final Answer:**
10	**2^10** Stack Hint: Powers of two	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

Unit 11 Time Trial: Combinatorics, Stats & Probability

Factorials, Permutations, Combinations & [SUM+] Accumulation • Focus: Probability Dice & Modular Spinner

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**5! (Factorial)** Stack Hint: x factorial key	Keystroke Plan:	**Final Answer:**
2	**6! (Factorial)** Stack Hint: x factorial key	Keystroke Plan:	**Final Answer:**
3	**Permutations P(5, 2) = 5! / 3!** Stack Hint: Stack permutation	Keystroke Plan:	**Final Answer:**
4	**Permutations P(6, 3) = 6! / 3!** Stack Hint: Stack permutation	Keystroke Plan:	**Final Answer:**
5	**Combinations C(5, 2) = 5! / (2! × 3!)** Stack Hint: n choose r	Keystroke Plan:	**Final Answer:**
6	**Combinations C(6, 2) = 6! / (2! × 4!)** Stack Hint: n choose r	Keystroke Plan:	**Final Answer:**
7	**Combinations C(7, 3) = 7! / (3! × 4!)** Stack Hint: n choose r	Keystroke Plan:	**Final Answer:**
8	**Series sum 1+2+3+4+5 via [SUM+]** Stack Hint: Statistical summation	Keystroke Plan:	**Final Answer:**
9	**4! (Factorial)** Stack Hint: x factorial key	Keystroke Plan:	**Final Answer:**
10	**Probability: 1 ÷ C(6, 2)** Stack Hint: Reciprocal combination	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

TIMED SPEED & FLUENCY SPRINT • SC-32 SCIENTIFIC

Unit 12 Time Trial: Digital Logic, Base Conversions & CS

HEX/DEC/OCT/BIN Radix Modes & Bitwise AND, OR, XOR Operations • Focus: Ten-Frame Tray & Counters

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**DEC 255 to HEX** Stack Hint: Radix mode cycle	Keystroke Plan:	**Final Answer:**
2	**HEX 10 to DEC** Stack Hint: Hex to decimal	Keystroke Plan:	**Final Answer:**
3	**DEC 15 to BIN** Stack Hint: Dec to binary	Keystroke Plan:	**Final Answer:**
4	**BIN 1010 to DEC** Stack Hint: Bin to decimal	Keystroke Plan:	**Final Answer:**
5	**Bitwise AND: 12 AND 10** Stack Hint: 1100 AND 1010 = 1000	Keystroke Plan:	**Final Answer:**
6	**Bitwise OR: 12 OR 10** Stack Hint: 1100 OR 1010 = 1110	Keystroke Plan:	**Final Answer:**
7	**Bitwise XOR: 12 XOR 10** Stack Hint: 1100 XOR 1010 = 0110	Keystroke Plan:	**Final Answer:**
8	**HEX 20 to DEC** Stack Hint: Hex 20 is 32	Keystroke Plan:	**Final Answer:**
9	**DEC 64 to OCT** Stack Hint: Dec 64 is octal 100	Keystroke Plan:	**Final Answer:**
10	**Bitwise AND: 255 AND 15** Stack Hint: Low nybble mask	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

TIMED SPEED & FLUENCY SPRINT • SC-32 SCIENTIFIC

Cumulative Trial C3: Grand RPN Invitational Championship

Master UIL-Style Time Trial across Units 1–12 (Complete Architecture) • Focus: All-Level StackCalc Mathematics Lab

MAX STACK DEPTH: 4 | **TIME LIMIT: 20 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ____:____ **Score: ____ / 50**

1	**Compound pipeline: 250 × (1 + .06/12)^24** Stack Hint: TVM compounding	Keystroke Plan:	**Final Answer:**
2	**Pythagorean distance: P1(-3, 5) to P2(5, -10)** Stack Hint: Distance pipeline triple	Keystroke Plan:	**Final Answer:**
3	**Polar vector: 12 cos(45°) + 5 sin(45°)** Stack Hint: Vector projection	Keystroke Plan:	**Final Answer:**
4	**Fractions: 3 1/3 ÷ 1 1/4 + 2/5** Stack Hint: Mixed operations	Keystroke Plan:	**Final Answer:**
5	**Combinations: C(8, 3) = 8! / (3! × 5!)** Stack Hint: Lottery combination	Keystroke Plan:	**Final Answer:**
6	**Exponential: 1500 × e^(0.05 × 10)** Stack Hint: Continuous growth	Keystroke Plan:	**Final Answer:**
7	**Postfix evaluation: ((35-7)×4)÷(14+2)** Stack Hint: Parenthesis elimination	Keystroke Plan:	**Final Answer:**
8	**Bitmasking: 192 AND 240 (Subnet mask)** Stack Hint: Network bitwise logic	Keystroke Plan:	**Final Answer:**
9	**Double & near-double sum: (6 ENTER +) + (8 ENTER** Stack Hint: Stack auto-lift chain	Keystroke Plan:	**Final Answer:**
10	**Radix conversion: DEC 4095 to HEX** Stack Hint: Maximum 12-bit hex	Keystroke Plan:	**Final Answer:**

UIL Calculator Competition Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this sprint. Round non-exact decimals to 4 significant digits.

Official Solutions & Keystrokes — Part 1

Verified answers and optimal RPN keystroke sequences for Part 1 (Trials 1–6 & Cumulative C1).

Unit 1 Time Trial: Step Counter & (Stack 1)

1. Ans: 3 — [1] [ENTER] [1] [+] [1] [+]
2. Ans: 7 — [5] [ENTER] [1] [+] [1] [+]
3. Ans: 4 — [1] [ENTER] [+] [+] [+]
4. Ans: 8 — [1] [ENTER] [+] [+] [+] [+]
5. Ans: 4 — [6] [ENTER] [1] [-] [1] [-]
6. Ans: 5 — [C] [1] [ENTER] [+] [+] [+]
7. Ans: 6 — [2] [ENTER] [2] [+] [2] [+]
8. Ans: 10 — [7] [ENTER] [1] [+] [1] [+]
9. Ans: 9 — [0] [ENTER] [9] [+]
10. Ans: 7 — [10] [ENTER] [1] [-] [1] [-]

Unit 2 Time Trial: Part-Part-Whol (Stack 2)

1. Ans: 7 — [4] [ENTER] [3] [+]
2. Ans: 10 — [5] [ENTER] [5] [+]
3. Ans: 3 — [9] [ENTER] [6] [-]
4. Ans: 6 — [10] [ENTER] [4] [-]
5. Ans: 15 — [8] [ENTER] [7] [+]
6. Ans: 6 — [14] [ENTER] [8] [-]
7. Ans: 13 — [6] [ENTER] [7] [+]
8. Ans: 9 — [18] [ENTER] [9] [-]
9. Ans: 20 — [12] [ENTER] [8] [+]
10. Ans: 14 — [25] [ENTER] [11] [-]

Unit 3 Time Trial: Ten-Frames & C (Stack 2)

1. Ans: 3 — [10] [ENTER] [7] [-]
2. Ans: 6 — [10] [ENTER] [4] [-]
3. Ans: 13 — [8] [ENTER] [2] [+] [3] [+]
4. Ans: 15 — [9] [ENTER] [1] [+] [5] [+]
5. Ans: 13 — [7] [ENTER] [3] [+] [3] [+]
6. Ans: 9 — [10] [ENTER] [1] [-]
7. Ans: 13 — [20] [ENTER] [7] [-]
8. Ans: 15 — [8] [ENTER] [2] [+] [5] [+]
9. Ans: 16 — [10] [ENTER] [6] [+]
10. Ans: 8 — [10] [ENTER] [2] [-]

Unit 4 Time Trial: Number Racks, (Stack 2)

1. Ans: 12 — [6] [ENTER] [+]
2. Ans: 16 — [8] [ENTER] [+]
3. Ans: 15 — [7] [ENTER] [+] [1] [+]
4. Ans: 13 — [6] [ENTER] [+] [1] [+]
5. Ans: 18 — [9] [ENTER] [+]
6. Ans: 17 — [8] [ENTER] [+] [1] [+]
7. Ans: 12 — [4] [ENTER] [3] [×]
8. Ans: 20 — [5] [ENTER] [4] [×]
9. Ans: 24 — [12] [ENTER] [+]
10. Ans: 19 — [9] [ENTER] [+] [1] [+]

Cumulative Trial C1: Primary Stac (Stack 2)

1. Ans: 7 — [1] [ENTER] [+] [+] [+] [+]
2. Ans: 16 — [9] [ENTER] [7] [+]
3. Ans: 9 — [17] [ENTER] [8] [-]
4. Ans: 7 — [10] [ENTER] [3] [-]
5. Ans: 28 — [14] [ENTER] [+]
6. Ans: 23 — [11] [ENTER] [+] [1] [+]
7. Ans: 17 — [9] [ENTER] [1] [+] [7] [+]
8. Ans: 30 — [5] [ENTER] [6] [×]
9. Ans: 18 — [30] [ENTER] [12] [-]
10. Ans: 21 — [8] [ENTER] [+] [5] [+]

Unit 5 Time Trial: 4-Register Sta (Stack 4)

1. Ans: 36 — [18] [ENTER] [6] [-] [3] [×]
2. Ans: 5 — [24] [ENTER] [16] [+] [8] [÷]
3. Ans: 5 — [45] [ENTER] [12] [ENTER] [3] [-
4. Ans: 69 — [15] [ENTER] [4] [×] [18] [ENTER
5. Ans: 4 — [50] [ENTER] [14] [-] [3] [ENTER
6. Ans: 28 — [12] [ENTER] [5] [×] [8] [ENTER]
7. Ans: 8 — [20] [ENTER] [8] [-] [4] [×]
8. Ans: 24 — [72] [ENTER] [4] [ENTER] [5] [+]
9. Ans: 32 — [100] [ENTER] [16] [ENTER] [5] [
10. Ans: 6 — [8] [ENTER] [4] [+] [9] [ENTER]

Unit 6 Time Trial: Fraction Studi (Stack 4)

1. Ans: 0.75 — [.1.2] [ENTER] [.1.4] [+]
2. Ans: 0.25 — [.3.4] [ENTER] [.1.2] [-]
3. Ans: 3.75 — [2.1.2] [ENTER] [1.1.4] [+]
4. Ans: 1.25 — [.4.5] [1/x]
5. Ans: 0.625 — [.3.8] [ENTER] [.1.4] [+]
6. Ans: 0.5 — [.2.3] [ENTER] [.3.4] [×]
7. Ans: 1.5 — [.3.4] [ENTER] [.1.2] [÷]
8. Ans: 6 — [1.1.2] [ENTER] [4] [×]
9. Ans: 2.25 — [5] [ENTER] [2.3.4] [-]
10. Ans: 6 — [.1.3] [ENTER] [.1.6] [+] [12] [

Scoring Adjudication

Accept equivalent rational fractions or 4-sig-fig decimals (e.g., 3/4 or 0.75). Deduct 2 points if superfluous [ENTER] presses overflowed the trial's Max Stack Depth.

Official Solutions & Keystrokes — Part 2

Verified answers and optimal RPN keystroke sequences for Part 2 (Trials 7–12 & Cumulative C2–C3).

Unit 7 Time Trial: Postfix Evalua (Stack 4)

1. Ans: 50 — [8] [ENTER] [2] [+] [9] [ENTER]
2. Ans: 30 — [14] [ENTER] [6] [-] [3] [×]
3. Ans: 20 — [30] [ENTER] [4] [ENTER] [5] [×]
4. Ans: 12 — [5] [ENTER] [3] [+] [8] [ENTER]
5. Ans: 40 — [18] [ENTER] [3] [÷] [4] [+]
6. Ans: 5 — [10] [ENTER] [5] [ENTER] [2] [×]
7. Ans: 70 — [25] [ENTER] [5] [ENTER] [3] [×]
8. Ans: 8 — [7] [ENTER] [3] [+] [4] [×]
9. Ans: 36 — [48] [ENTER] [2] [ENTER] [6] [+]
10. Ans: 30 — [16] [ENTER] [4] [ENTER] [2] [+]

Unit 8 Time Trial: Tactile Algebr (Stack 4)

1. Ans: -2 — [1] [ENTER] [5] [ENTER] [6] [POL
2. Ans: -3 — [1] [ENTER] [7] [ENTER] [12] [PO
3. Ans: -3 — [1] [ENTER] [8] [ENTER] [15] [PO
4. Ans: 3 — [1] [ENTER] [1] [+/-] [ENTER] [6
5. Ans: 2 — [1] [ENTER] [5] [+/-] [ENTER] [6
6. Ans: 3 — [1] [ENTER] [0] [ENTER] [9] [+/-
7. Ans: 4 — [1] [ENTER] [0] [ENTER] [16] [+/
8. Ans: 25 — [3] [ENTER] [x²] [2] [×] [3]
9. Ans: -0.5 — [2] [ENTER] [5] [ENTER] [2] [POL
10. Ans: -5 — [1] [ENTER] [10] [ENTER] [25] [P

Cumulative Trial C2: Intermediate (Stack 4)

1. Ans: 15 — [32] [ENTER] [14] [-] [5] [×]
2. Ans: 0.875 — [.1.2] [ENTER] [.1.4] [×] [.3.4]
3. Ans: 10 — [9] [ENTER] [3] [-] [x²] [12]
4. Ans: 1.75 — [3.1.2] [ENTER] [1.3.4] [-]
5. Ans: 10 — [15] [ENTER] [5] [+] [8] [ENTER]
6. Ans: 27 — [18] [ENTER] [+] [9] [-]
7. Ans: 3 — [10] [ENTER] [2] [+/-] [-] [3]
8. Ans: 32 — [80] [ENTER] [15] [ENTER] [4] [×
9. Ans: 2.4 — [.1.4] [ENTER] [.1.6] [+] [1/x]
10. Ans: 20 — [12] [x²] [16] [x²] [+] [√▣]

Unit 9 Time Trial: Trigonometry, (Stack 4)

1. Ans: 0.5 — [30] [SIN]
2. Ans: 0.5 — [60] [COS]
3. Ans: 1 — [45] [TAN]
4. Ans: 5 — [10] [ENTER] [30] [SIN] [×]
5. Ans: 6 — [12] [ENTER] [60] [COS] [×]
6. Ans: 10 — [10] [ENTER] [0] [yellow] [->y,x]
7. Ans: 5 — [3] [ENTER] [4] [yellow] [->theta,r]
8. Ans: 13 — [5] [ENTER] [12] [yellow] [->theta,r]
9. Ans: 10 — [8] [ENTER] [6] [yellow] [->theta,r]
10. Ans: 30 — [.5] [yellow] [ASIN]

Unit 10 Time Trial: Exponential G (Stack 4)

1. Ans: 32 — [2] [ENTER] [5] [y^x]
2. Ans: 81 — [3] [ENTER] [4] [y^x]
3. Ans: 1 — [1] [e^x] [LN]
4. Ans: 1 — [0] [e^x]
5. Ans: 3 — [1000] [yellow] [LOG]
6. Ans: 1102.5 — [1000] [ENTER] [1.05] [ENTER] [2
7. Ans: 9.9021 — [2] [LN] [.07] [÷]
8. Ans: 12.5 — [100] [ENTER] [.5] [ENTER] [3] [
9. Ans: 610.7014 — [.04] [ENTER] [5] [×] [e^x] [500]
10. Ans: 1024 — [2] [ENTER] [10] [y^x]

Unit 11 Time Trial: Combinatorics (Stack 4)

1. Ans: 120 — [5] [yellow] [x!]
2. Ans: 720 — [6] [yellow] [x!]
3. Ans: 20 — [5] [yellow] [x!] [3] [yellow] [
4. Ans: 120 — [6] [yellow] [x!] [3] [yellow] [
5. Ans: 10 — [5] [yellow] [x!] [2] [yellow] [
6. Ans: 15 — [6] [yellow] [x!] [2] [yellow] [
7. Ans: 35 — [7] [yellow] [x!] [3] [yellow] [
8. Ans: 15 — [1] [SUM+] [2] [SUM+] [3] [SUM+]
9. Ans: 24 — [4] [yellow] [x!]
10. Ans: 0.0667 — [6] [yellow] [x!] [2] [yellow] [

Unit 12 Time Trial: Digital Logic (Stack 4)

1. Ans: FF — [255] [yellow] [BASE] [HEX]
2. Ans: 16 — [HEX] [10] [yellow] [BASE] [DEC]
3. Ans: 1111 — [15] [yellow] [BASE] [BIN]
4. Ans: 10 — [BIN] [1010] [yellow] [BASE] [DE
5. Ans: 8 — [12] [ENTER] [10] [AND]
6. Ans: 14 — [12] [ENTER] [10] [OR]
7. Ans: 6 — [12] [ENTER] [10] [XOR]
8. Ans: 32 — [HEX] [20] [yellow] [BASE] [DEC]
9. Ans: 100 — [64] [yellow] [BASE] [OCT]
10. Ans: 15 — [255] [ENTER] [15] [AND]

Cumulative Trial C3: Grand RPN In (Stack 4)

1. Ans: 281.79 — [.06] [ENTER] [12] [÷] [1] [+]
2. Ans: 17 — [5] [ENTER] [3] [+/-] [-] [x²]
3. Ans: 12.0208 — [12] [ENTER] [45] [COS] [×] [5]
4. Ans: 3.0667 — [3.1.3] [ENTER] [1.1.4] [÷] [.2.
5. Ans: 56 — [8] [yellow] [x!] [3] [yellow] [
6. Ans: 2473.08 — [.05] [ENTER] [10] [×] [e^x] [150
7. Ans: 7 — [35] [ENTER] [7] [-] [4] [×]
8. Ans: 192 — [192] [ENTER] [240] [AND]
9. Ans: 29 — [6] [ENTER] [+] [8] [ENTER] [+]
10. Ans: FFF — [4095] [yellow] [BASE] [HEX]

Scoring Adjudication

Accept equivalent rational fractions or 4-sig-fig decimals (e.g., 3/4 or 0.75). Deduct 2 points if superfluous [ENTER] presses overflowed the trial's Max Stack Depth.

Speed Calculation & Formula Reference

Keystroke-saving RPN techniques from Thinking in the Stack (Chapter 14) with authentic keycap sequences.

1. START AT THE DEEPEST INNER EXPRESSION (ZERO PARENTHESES)

Evaluate the most deeply nested quantity first so intermediate results wait in Register Y. Example: For 100 ÷ (3 + 4×5), compute 4×5 + 3 = 23 first, then type 100 and swap:

Keycap Sequence: [4] [ENTER] [5] [×] [3] [+] [1] [0] [0] [↑↓] [÷]

2. INSTANT DOUBLING & SQUARING ([ENTER] [+] AND [ENTER] [×])

Pressing [ENTER] duplicates Register X into Register Y without retyping digits. To double 18 and add 5 in 5 keystrokes:

Keycap Sequence: [1] [8] [ENTER] [+] [5] [+]

3. NON-COMMUTATIVE SWAP (X SWAP Y) & RECIPROCAL ([1/x])

If you computed the denominator D first and then typed numerator N into X, press [x swap y] [÷]—or invert D in place with [1/x] and multiply:

Keycap Sequence: [3] [ENTER] [6] [+] [x^{-1}] [4] [5] [×]

4. PARALLEL RESISTANCE & HARMONIC SUMS (AUTOMATIC STACK LIFT)

To compute 1 / (1/4 + 1/6), never use memory registers—automatic stack lift pushes the first reciprocal into Y while the second is computed:

Keycap Sequence: [4] [x^{-1}] [6] [x^{-1}] [+] [x^{-1}]

5. VECTOR MAGNITUDE & POLAR CONVERSION (TO POLAR)

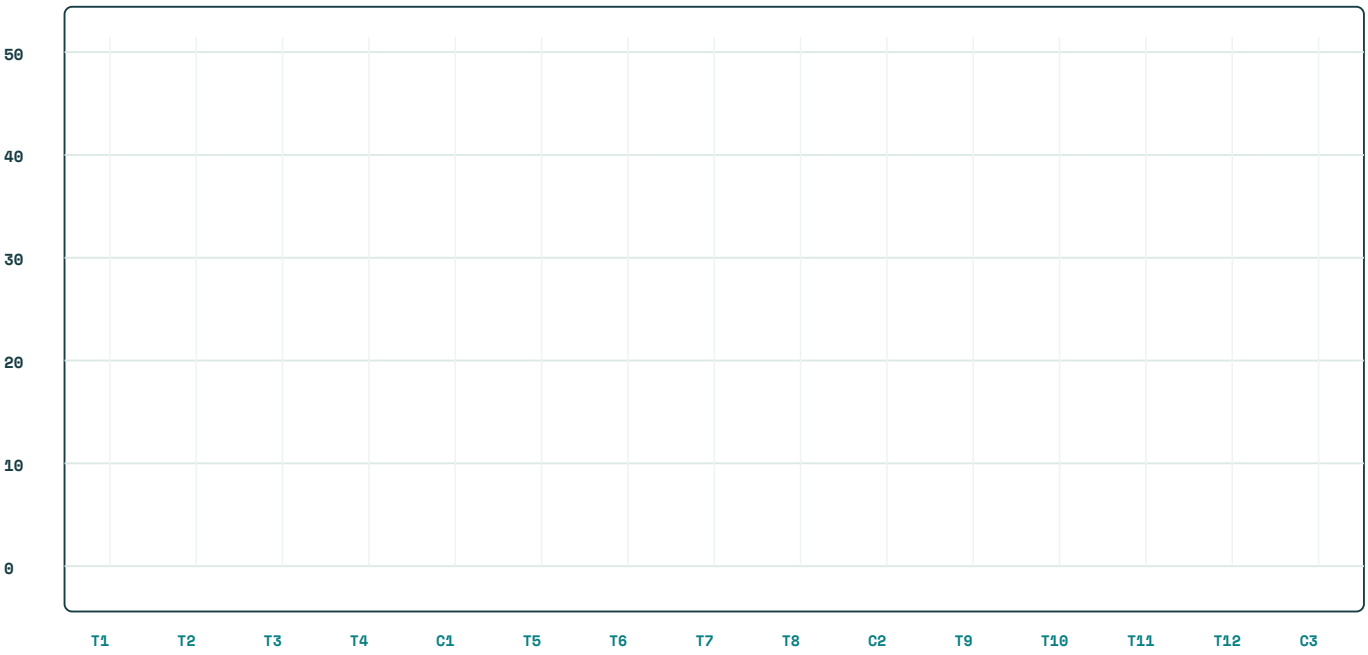
To find the hypotenuse or magnitude of components (x=5, y=12) in a single operation on the SC-32:

Keycap Sequence: [1] [2] [ENTER] [5] [↶] [▸θ,r]

Competitor Progress Graph & Certificate

Plot your score across all 15 timed trials and record your Stack-Depth Mastery certification.

1. 15-TRIAL SCORE PROGRESSION GRAPH (PLOT YOUR SCORE FROM 0 TO 50 POINTS)



OFFICIAL CERTIFICATE OF RPN STACK PROFICIENCY

This certifies that: _____

has completed the 15 timed sprints in the StackCalc Stack Speed & Fluency Trials, demonstrating numerical accuracy, inside-out operand sequencing, and strict 4-register (T, Z, Y, X) stack depth discipline.

[ ] Level 1 Stack Badge (Single Register X) **[ ] Level 2 Stack Badge (Registers X & Y)**

[ ] Level 4 Master Badge (Full T-Z-Y-X Stack) **Highest Trial Score: _____ / 50 pts**

Coach / Teacher Signature: _____ Date: _____